



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Advanced Level

COMPUTER SCIENCE
PAPER 1

6023/1

NOVEMBER 2021 SESSION

3 hours

Additional materials:
Answer paper

TIME 3 hours

INSTRUCTIONS TO CANDIDATES

Write your name, centre number and candidate number in the spaces provided on the answer paper/answer booklet.

Answer **all** questions.

Write your answers on the separate answer paper provided.

If you use more than one sheet of paper, fasten the sheets together.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question.

You are reminded of the need for good English and clear presentation in your answers.

This question paper consists of 5 printed pages and 3 blank pages.
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- 1 (a) Illustrate using diagrams the difference between the Von Neumann Architecture and the Harvard Architecture. [6]
- (b) Name any **four** components of the CPU. * [4]✓
- 2 (a) Three descriptions and two types of processor are given below. Link each description to the appropriate type of processor.

Description	Types of Processor
Makes extensive use of general purpose registers	RISC
Many addressing modes are available	CISC
Has a simplified set of instructions	

[3]

- (b) In a RISC processor three instructions (A followed by B, followed by C) are processed using pipelining.

The following table shows the five stages that occur when instructions are fetched and executed.

Stage	Time Interval								
	1	2	3	4	5	6	7	8	9
Fetch Instruction	A								
Decode Instruction									
Execute Instruction									
Access operand in Memory									
Write result to register									

- (i) The 'A' in the table indicates that instruction A has been fetched in time interval 1. Copy and complete the table to show the interval in which each stage of each instruction (A, B and C) is carried out. [4]

- (ii) The completed table shows how pipelining allows instructions to be carried out more rapidly. Each time interval represents one clock cycle.

Calculate the number of clock cycles saved by the use of
 ✕ pipelining in the above example. Show your working. [3]

- 3 (a) Convert the following hexadecimal numbers to their decimal equivalent. Show the calculations.

(i) 2 0 B 3 [3]

(ii) 1 2 A E 5 ✕ [3]

- (b) Binary Coded Decimal (BCD) is another way of representing numbers.

(i) Write the number 359 in BCD form. [2]

(ii) Describe the use of BCD number representation, with an example. [2] ✕

- 4 In a particular Computer System, real numbers are stored using floating-point representation with

8 bits for the mantissa, followed by 8 bits for the exponent

Two's complement form is used for both mantissa and exponent.

- (a) A real number is stored as the following two bytes.

Mantissa	Exponent
0 0 1 0 1 0 0 0	0 0 0 0 0 0 1 1

Calculate the denary value of this number. Show your working. [3]

- (b) Explain why the floating-point number in part (a) is not normalised. [2]

- 5 Examine any **five** factors why e-business is needed. [10]

- 6 Cloud computing is now used mostly by individuals and organisations.

(a) Define *cloud computing*. [2]

(b) Give any **two** cloud services that are commonly used. ✕ [2]

(c) List any **three** cloud service models and their purposes. ✕ [6]

7 The world over, devices access to the Word Wide Web (www) content using IP addressing.

(a) Expand the acronym IP. [1]

(b) Copy and complete the table below, indicating for each IP address whether it is valid or invalid and give a suitable reason.

address	denary /hexadecimal	valid or invalid	reason
3.2A.6AA.BBBB	hexadecimal		
2.0.255.1	denary		
6.0.257.6	denary		
A.78.F4.JB	hexadecimal		

[6]*

(c) Distinguish between *public* and *private IP addresses*. [3]

8 (a) Define the following terms.

(i) Firewall [2]

(ii) Authentication [2]

(b) Distinguish between *data integrity* and *data security*. [2]

(c) Data integrity is required at the input stage and also during transfer of data.

(i) Suggest any **two** ways of maintaining data integrity at the input stage. Use examples to help explain your answer. [2]

(ii) Show any **two** ways of maintaining data integrity during data transmission. Use examples to explain your answer. [2]

9 (a) Define the terms *sorting* and *searching*. [2]

(b) Analyse the *bubble sort method* and the *quick sort method*. [6]

(c) Produce the binary search method showing the algorithm syntax. [7] *

- 10 (a) A relational database management system (RDBMS) implements features of the relational model. Outline any **three** components of the (RDBMS) and explain each. [6]
- (b) Outline **four** different data types available in SQL. [4]